

Synopsis

Project: **Data Analytics on Global Tuberculosis Burden Across Countries.**

Team Members:

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Introduction:

Tuberculosis (TB) remains one of the top infectious diseases globally, despite being preventable and curable. According to the World Health Organization (WHO), millions of new cases occur every year, with varying prevalence across regions. Data analytics can reveal patterns in TB incidence, mortality, and treatment success rates—helping policymakers and health organizations better target interventions and allocate resources.

Problem Statement:

Although WHO and other global health bodies collect TB statistics annually, the data is often under-utilized for comparative analysis and actionable insights. There is a need to analyze country-wise and regional TB burdens to identify high-risk areas, trends over time, and the impact of healthcare efforts.

Objective:

- To analyze the **country-wise TB burden** using WHO datasets.
- To study **trends in TB incidence, prevalence, and mortality** over time.
- To compare **TB cases across regions (Africa, Asia, Europe, etc.)**.
- To identify **countries with the highest TB rates per 100,000 population**.
- To explore **correlations between TB burden and socioeconomic indicators** (like GDP, healthcare spending).

Dataset Description:

Source: World Health Organization

Format: CSV

Size: 100,000+ observations

Key Fields:

- Country Name
- Year
- TB Incidence (per 100,000)
- TB Mortality (per 100,000)
- Population
- HIV-TB Co-infection rates
- Treatment Success Rate

Tools & Technologies:

Depending on the depth of analysis, the following tools may be used:

- **Excel / Power BI / Tableau** – Visualization and dashboards
- **Python (Pandas, Matplotlib/Plotly)** – Data preprocessing and analysis
- **SQL (optional)** – Querying the dataset

Methodology:

- **Data Collection** – Obtain TB dataset from WHO portal
- **Data Cleaning** – Handle missing data, unify formats, remove duplicates
- **Exploratory Data Analysis (EDA)**
 - Identify top 10 countries by TB burden
 - Analyze year-wise trends in incidence and mortality
 - Compare TB rates across continents
 - Visualize correlation between TB rates and socioeconomic factors
- **Visualization & Insights** – Create charts, heatmaps, dashboards
- **Interpretation** – Summarize findings, discuss potential causes and implications

Expected Outcomes:

- Identification of countries and regions with the highest TB burden
- Trend analysis of TB incidence and mortality over time
- Insights into how economic or healthcare indicators relate to TB control
- Data-driven recommendations for public health planning

Conclusion:

This project will provide an analytical view of tuberculosis trends worldwide. By leveraging WHO data and visualization tools, it will uncover valuable patterns to support global health decision-making and disease prevention strategies.